

The Influence of Intonation Changes Based on Heart Rate on Empathy Formation with a Chatbot

Yuuta Noguchi¹ and Makoto Nokata¹

Abstract—This study proposes a Heart Rate Feedback (HFB) system where a chatbot’s speech intonation is adjusted based on the user’s heart rate to foster cognitive empathy. In an experiment with 12 subjects (9 valid), this method was compared against a fixed-intonation condition and a sine-wave condition. Heart rate variability analysis indicated that the fixed-intonation condition induced higher sympathetic nervous activity. Furthermore, subjective evaluations showed that the HFB condition significantly reduced user stress and increased the feeling of being listened to. These findings suggest that reflecting a user’s internal biological state in a robot’s expression, even in a simple manner, can enhance the quality of interaction and promote a sense of being understood.

1. INTRODUCTION

In recent years, as robots and machines increasingly interact with people, there has been a growing focus on communication with them, and further, on the empathy that can arise between robots and humans [1]. Humans acquire information through their five senses to communicate with others, and this sensory information is necessary for empathy to develop. However, robots, unlike humans, can acquire information through means other than the five senses, with biological information being one such example.

Furthermore, merely acquiring information is not enough to establish empathy. A mutual relationship is necessary, and adapting to the other’s state is a crucial part of it. Empathy is broadly divided into “cognitive empathy” and “affective empathy.” The act of recognizing and adapting to another’s state is considered “cognitive empathy” [2].

Previous efforts have been made to foster empathy in human-robot communication using information beyond the five senses. Nishimura et al. aimed to enhance “affective empathy” by synchronizing a robot’s vocal elements with the user’s, finding that the synchronized robot was perceived as more friendly and emotionally connected [3]. Jonell successfully improved the subjective quality of conversations by utilizing biological signals such as skin temperature and respiratory rate [4]. Saffaryazdi et al. used biological signals (heart rate, EEG) for emotion recognition in a digital human, enabling it to provide empathetic responses [5]. However, these studies require a “learning machine” for emotions from biological data. We argue that this is not a genuine approach of using biological signals directly to alter a robot’s approach to a human.

Therefore, the objective of this study is to determine whether changing intonation based on heart rate feedback

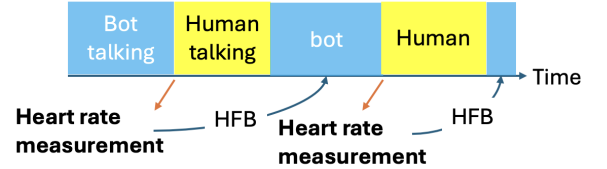


Fig. 1. Flow of Heart Rate Feedback (HFB). The system captures the user’s heart rate just as they begin to speak and uses it to adjust the bot’s next utterance.

enhances empathy and reduces stress. We evaluate this using subjective assessments and physiological data to measure changes in autonomic nervous system activity, empathy levels, and stress levels.

2. PROPOSED SYSTEM

In this study, we focus on eliciting cognitive empathy from a robot by emphasizing two essential processes: (1) recognizing the user’s state, and (2) adapting to that state. Based on this concept, we propose a Heart Rate Feedback (HFB) method, in which the robot acquires the user’s heart rate in real-time (to recognize the state) and dynamically adjusts the chatbot’s vocal intonation based on the heart rate fluctuations (to adapt to the state).

The procedure of the HFB method employed in this study is illustrated in Fig. 1.

During conversation, a listener processes the speaker’s message and mentally prepares their own response. Once this internal processing reaches a certain level, the listener begins to speak. We assume this process applies similarly to conversations between a human and a robot. In this study, we hypothesize that the user’s internal state immediately prior to speaking strongly reflects their reaction to the ongoing dialogue. Accordingly, we use the heart rate measured at this moment as the input reference for the HFB system.

We then examined how to convey this acquired physiological information in a way that effectively demonstrates the robot’s adaptation to the user. Information can be conveyed through verbal or non-verbal channels. To express the process of “understanding the other”—a central aspect of cognitive empathy—we focus on non-verbal communication, which is often more natural and implicit than modifying verbal content. Among various non-verbal cues such as facial expressions, gestures, and vocal tone, we emphasize intonation, which is both emotionally expressive and modifiable in real time within spoken interactions.

Specifically, the chatbot’s vocal intonation is adjusted based on the deviation of the user’s heart rate from their

¹Yuuta Noguchi and Makoto Nokata are with Ritsumeikan University, 1-1-1 Noji-higashi, Kusatsu, Shiga, 525-8577, Japan

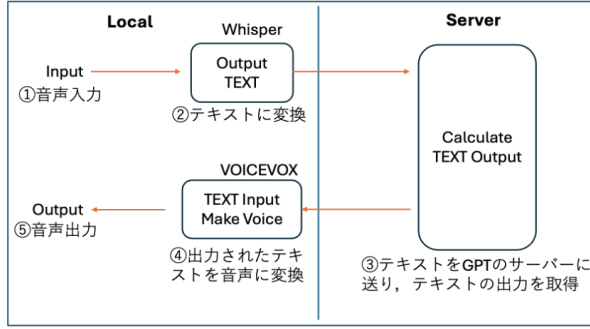


Fig. 2. Flow of the Conversation System.

baseline. This adjustment is designed to create the impression that the robot is attuned to the user’s internal state, thereby fostering empathy and improving the quality of human-robot interaction.

3. EXPERIMENT

3.1 Conversation System

The conversation system architecture is shown in Fig. 2. (1) The user’s speech is recorded. Recording starts on voice detection and ends 1.6 seconds after the user stops speaking. (2) The audio is transcribed to text using OpenAI’s Whisper. (3) The text is sent to OpenAI’s GPT-4 to generate a response. (4) The response text is synthesized into speech using VOICEVOX. (5) The synthesized audio is played. The average response time from the end of the user’s utterance to the bot’s reply was 4.57 seconds.

3.2 Experimental Details

Nine male and three female subjects aged 20 to 27 (9 valid datasets) participated in a 7-minute free-form conversation on the topic of “hobbies.” The first and last minutes were excluded, and the middle 5 minutes were used for analysis. Each subject experienced all three of the following conditions in a randomized (counterbalanced) order.

- **Fixed Condition:** The chatbot’s intonation was fixed at a constant value (1.0).
- **Adjusted Condition (HFB):** The bot’s intonation was adjusted in real-time. For every ± 1 BPM deviation from the user’s baseline resting heart rate, the intonation was adjusted by ± 0.1 (range: 0.0–2.0).
- **Sine Wave Condition:** The chatbot’s intonation was varied along a sine wave in 0.2 increments. The change was triggered at the same moment as in the adjusted condition (when the user started speaking).

3.3 Evaluation Method

We evaluated the system’s impact using physiological and subjective measures. Autonomic nervous system activity, which reflects arousal states like “excitement” or “stress,” was assessed using two heart rate variability (HRV) metrics:

- **RMSSD (Root Mean Square of Successive Differences):** Primarily reflects parasympathetic activity. Lower values indicate sympathetic dominance (arousal/stress). Calculated over 30-second windows.

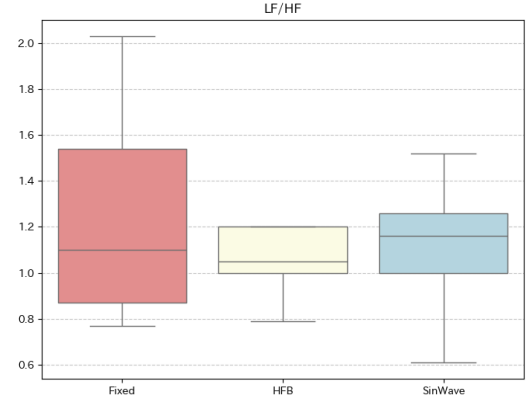


Fig. 3. Boxplot of LF/HF ratio over 5 minutes. The Fixed condition shows the highest median value, indicating greater sympathetic activity.

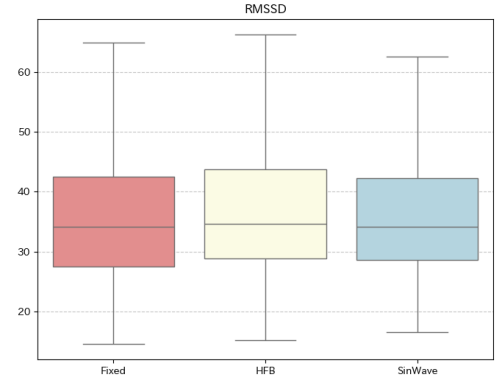


Fig. 4. Boxplot of RMSSD values. The Fixed condition shows the lowest median RMSSD, corroborating the finding of increased sympathetic arousal.

- **LF/HF:** The ratio of low-frequency (LF) to high-frequency (HF) power. It represents the balance of the autonomic nervous system. Higher values indicate sympathetic dominance. Calculated over the entire 5-minute conversation.

Since physiological arousal can be positive or negative, we administered a post-session questionnaire to gather subjective ratings for “Empathy,” “Stress,” and “Did the bot listen to you?”. A Visual Analog Scale (VAS) was used, with a maximum scale of 183 mm.

3.4 Results

The median LF/HF value for the entire 5-minute interaction was highest in the Fixed condition (1.19), followed by the Sine Wave (1.09) and Adjusted (1.04) conditions, as shown in Fig. 3. The RMSSD results are shown in Fig. 4. An ANOVA on RMSSD data revealed a significant main effect of the condition ($F(2, 2852) = 17.04, p < .001$). The Fixed condition had a significantly lower RMSSD compared to the Adjusted and Sine Wave conditions.

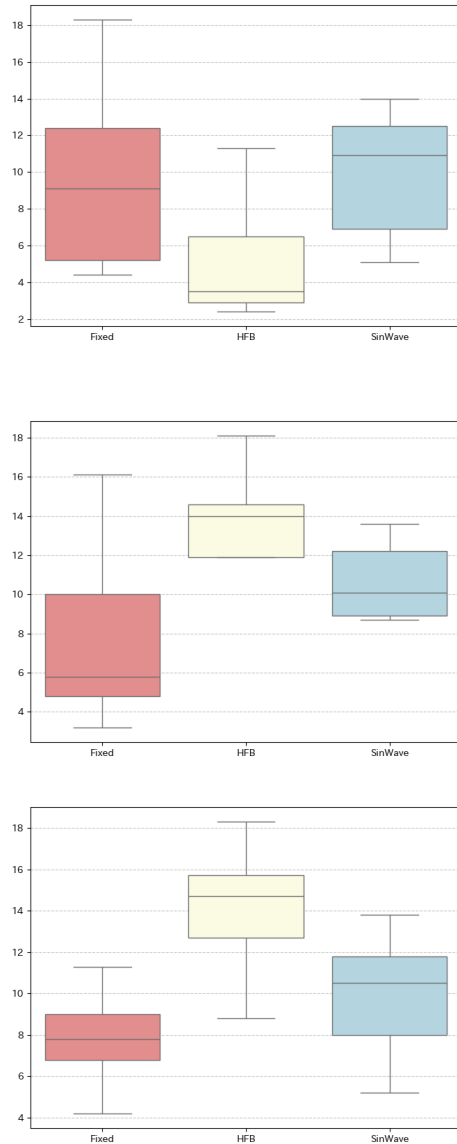


Fig. 5. Questionnaire Results (VAS scores in mm). Top: Stress. Middle: Empathy. Bottom: "Did the bot listen to you?". The Adjusted condition performed best across all metrics.

Subjective evaluation results are shown in Fig. 5. A Friedman test showed significant differences among conditions for "Stress" and "Did the bot listen to you?". The Adjusted condition was rated as significantly less stressful and significantly higher on being listened to compared to the other conditions. While the "Empathy" score was highest in the Adjusted condition, the difference was not statistically significant.

4. DISCUSSION

The results present a consistent trend between physiological and subjective data. The physiological metrics (lower RMSSD, higher LF/HF) indicate that the subjects' sympathetic nervous systems were most active during the **Fixed condition**. This suggests that the monotonous, unchanging

interaction may have acted as a stressor or required more cognitive effort to maintain engagement, leading to increased physiological arousal.

In contrast, the subjective evaluations clearly favored the **Adjusted condition**. This demonstrates the effectiveness of adapting to the user's physical state. Even though it was a simple rule-based system, the fact that the bot's intonation was linked to their own heart rate likely gave subjects a sense that their internal state was being understood and considered. This feeling of "being understood"—a form of cognitive empathy—may have fostered a sense of security, which in turn suppressed the physiological stress response and led to higher subjective ratings of comfort.

Therefore, we conclude that the proposed HFB method is effective in improving the subjective quality of the interaction, leading to a reduction in both physiological and psychological stress.

5. CONCLUSION

This study verified that a chatbot adjusting its vocal intonation based on the user's heart rate positively impacts the interaction experience. The proposed Heart Rate Feedback (HFB) system was shown to significantly reduce subjective stress and increase the feeling of "being listened to" compared to a fixed-intonation bot. Physiological data indicated that a monotonous interaction is more arousing/stressful than an adaptive one. This suggests that non-verbal communication where a robot adapts to a user's physiological state, even in a simple way, is effective in promoting a sense of being understood, which can improve the quality of human-robot interaction.

Future work includes validating these findings with a more diverse group of participants (age, gender). Technical improvements, such as reducing system latency, are also necessary. We also plan to explore different feedback approaches. Through new research into how robots can acquire empathy, we aim to build relationships where robots and humans can mutually support each other.

References

- [1] C. Bartneck, T. Belpaeme, F. Eyssel, T. Kanda, M. Keijsers, and S. Šabanović, "Human-Robot Interaction: An Introduction," 2nd ed. Cambridge: Cambridge University Press, 2024.
- [2] J. Decety and P. L. Jackson, "The functional architecture of human empathy," *Behavioral and Cognitive Neuroscience Reviews*, vol. 3, no. 2, pp. 71–100, 2004.
- [3] S. Nishimura, T. Nakamura, W. Sato, M. Kanbara, Y. Fujimoto, H. Kato, and N. Hagita, "Vocal Synchrony of Robots Boosts Positive Affective Empathy," *Applied Sciences*, vol. 11, no. 6, p. 2502, 2021.
- [4] P. Jonell, "Using Social and Physiological Signals for User Adaptation in Conversational Agents," in *Proc. of AAMAS '19*, May 13–17, Montréal, Canada, pp. 2109–2111, 2019.
- [5] N. Saffaryazdi, T. S. Gunasekaran, K. Laveys, E. Broadbent, and M. Billingham, "Empathetic Conversational Agents: Utilizing Neural and Physiological Signals for Enhanced Empathetic Interactions," *arXiv preprint arXiv:2501.08393*, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2501.08393>